ROBO MATES

KHANISHK — MONTH 1 STUDY PLAN

Robotics Engineering Foundation | 8 November – 7 December 2026

Month 1 target: Build strong beginner foundations in Python, Git and Linux. Finish with a working command-line Robot Mission Logger. This month is deliberately software-first; no expensive robotics hardware is required yet.

Daily commitment: Approximately 4–5 hours on study days. Use focused blocks: 60 min learning, 60 min guided practice, 90 min independent exercises, 60 min project work and 30 min review. Adjust on review days. Rest days are not missed days.

Month-end deliverables: Python exercise folder; Git repository; GitHub repository with README; Robot Mission Logger v1; ten-test checklist; short demo; list of gaps to address in Month 2.

1. Equipment and setup checklist

Item	Need now?	Recommendation / purpose	Estimated cost
Laptop	Already have	Your existing Windows laptop is sufficient for Month 1.	■0
Internet access	Yes	For documentation, videos, GitHub and downloads.	Existing connection
Python 3	Yes	Free language runtime from python.org.	Free
VS Code or PyCharm Community	Yes	Choose one IDE; do not install multiple IDEs unnecessarily.	Free
Git	Yes	Version control; install Git for Windows.	Free
GitHub account	Yes	Store your learning repository; use a strong password and 2FA.	Free tier
Notebook and pen	Recommended	Record concepts, errors, pseudocode and daily reflections.	■100–■250 optional
USB drive / cloud backup	Optional	Backup code and notes; GitHub also provides remote backup.	Optional
Arduino / ESP32 / sensors / motors	Not this month	Do not buy yet. First complete programming and project foundations; hardware selection comes after the Month 2–3 plan.	■0 this month

2. Software installation checklist

• Install Python 3 from https://www.python.org/downloads/ . During Windows installation, enable the option to add Python to PATH if offered.

• Install one IDE: VS Code (https://code.visualstudio.com/) or PyCharm Community (https://www.jetbrains.com/pycharm/download/).

• Install Git for Windows from https://git-scm.com/download/win .

• Create a GitHub account at https://github.com/ . Enable two-factor authentication.

• Create a folder such as Documents/ROBO_MATES/Month_01_Python. Keep code, notes, screenshots and project files organized.

• Verify in PowerShell: python --version, git --version. If the python command is unavailable, check the Python installer PATH option or use the Python launcher (py --version).

3. Where to study — reliable learning sources

Python official tutorial — https://docs.python.org/3/tutorial/
Main reference for syntax, data structures, modules, files and classes. Read the matching section after the daily lesson.

Python beginner documentation — https://www.python.org/about/gettingstarted/
Beginner-friendly entry point and installation guidance.

CS50P — Introduction to Programming with Python — https://cs50.harvard.edu/python/
Free course materials and problem sets. Use the relevant early lessons; do not try to finish the entire course in one month.

Git book — Pro Git — https://git-scm.com/book/en/v2
Free reference. Focus on Getting Started and Git Basics.

GitHub Skills — https://skills.github.com/
Interactive practice for repositories, commits and collaboration.

PowerShell documentation — https://learn.microsoft.com/powershell/
Reference for navigating folders and running commands on Windows.

Python practice — https://exercism.org/tracks/python
Optional practice exercises; use after learning each topic.

4. Complete day-by-day plan

Each day specifies the topic, exact subtopics, practical work and evidence to save. Planned hours are targets, not a reason to rush. Finish the exercise and understand it before moving forward.

DAY 1 — Sunday, 08 November 2026 | LEARNING | 4 HOURS
Topic: Setup Python and tools
Study: Python 3 installation; IDE setup; interpreter; terminal; project folder structure.
Do: Install/verify Python. Run python --version. Create hello.py and run it in IDE and PowerShell.
Save: Working environment; hello.py; screenshot of version/output.

DAY 2 — Monday, 09 November 2026 | LEARNING | 4 HOURS
Topic: Your first Python program
Study: print(); comments; strings; escape characters; script structure.
Do: Write a program that prints a short profile and five facts about a robot you want to build. Add comments and run it twice.
Save: intro.py with readable comments.

DAY 3 — Tuesday, 10 November 2026 | LEARNING | 4 HOURS
Topic: Variables and data types
Study: Variable assignment; naming; int, float, str, bool; type().
Do: Create variables for robot name, wheel count, battery voltage and simulation flag. Print values and types.
Save: variables.py and short explanation.

DAY 4 — Wednesday, 11 November 2026 | LEARNING | 4 HOURS
Topic: Input and conversion
Study: input(); int(); float(); str(); f-strings; invalid input.
Do: Ask for a name and two numeric measurements; convert them and display a formatted summary. Test invalid input.
Save: measurement_input.py with valid/invalid test notes.

DAY 5 — Thursday, 12 November 2026 | LEARNING | 4 HOURS
Topic: Operators and calculations
Study: Arithmetic; precedence; comparisons; boolean expressions.
Do: Calculate wheel circumference and distance travelled from rotations using sample values. Test three cases and explain units.
Save: wheel_distance.py and three test cases.

DAY 6 — Friday, 13 November 2026 | WEEKLY ASSESSMENT | 3.5 HOURS
Topic: Week 1 review
Study: Python setup, output, variables, types, input and operators.
Do: Take a 15-question self-quiz. Rebuild two exercises without notes. Fix one bug in an earlier script.
Save: Quiz score, corrected scripts and learning note.

DAY 7 — Saturday, 14 November 2026 | REST / REVIEW | 0 HOURS
Topic: Rest and setup review
Study: No new topic. File organization and recovery.
Do: Optional 20-minute recap. Back up Month1 folder. No mandatory study hours.
Save: Clean folder structure.

DAY 8 — Sunday, 15 November 2026 | LEARNING | 4 HOURS
Topic: Conditional statements
Study: if, elif, else; indentation; nested conditions; and/or/not.
Do: Build a battery-status checker using clearly labelled illustrative thresholds. Test at least six values.
Save: battery_status.py and test table.

DAY 9 — Monday, 16 November 2026 | LEARNING | 4 HOURS
Topic: For loops
Study: range(); iteration; counters; accumulation; loop patterns.
Do: Print 1–20; create a multiplication table; calculate sum and average of a list of mock sensor readings.
Save: for_loops.py with three exercises.

DAY 10 — Tuesday, 17 November 2026 | LEARNING | 4 HOURS
Topic: While loops
Study: while conditions; counters; break; continue; infinite-loop prevention.
Do: Build a command-line menu that repeats until Exit is selected. Test invalid choices and exit behavior.
Save: while_menu.py.

DAY 11 — Wednesday, 18 November 2026 | LEARNING | 4 HOURS
Topic: Strings
Study: Indexing; slicing; len; strip; split; replace; lower/upper; f-strings.
Do: Parse mock sensor text such as TEMP: 28.5 C. Handle missing or malformed values safely.
Save: string_parser.py and five test cases.

DAY 12 — Thursday, 19 November 2026 | LEARNING | 4 HOURS
Topic: Lists and tuples
Study: Create, index, update, append, remove, iteration; tuple basics.
Do: Store five mock distance readings; calculate minimum, maximum and average; explain the mock data.
Save: sensor_readings.py and output.

DAY 13 — Friday, 20 November 2026 | LEARNING | 4 HOURS
Topic: Functions
Study: def; parameters; return; scope; decomposition; docstrings.
Do: Refactor sensor calculations into reusable validation, average, minimum and maximum functions.
Save: functions_sensor.py.

DAY 14 — Saturday, 21 November 2026 | WEEKLY ASSESSMENT | 3.5 HOURS
Topic: Week 2 review
Study: Conditionals, loops, strings, lists and functions.
Do: Complete a 20-question quiz. Solve two small problems without copying. Explain one bug and its fix.
Save: Quiz result and revised code.

DAY 15 — Sunday, 22 November 2026 | REST / REVIEW | 0 HOURS
Topic: Rest and review
Study: No new topic. Review weak areas.
Do: Optional light practice; write down five questions to resolve next week.
Save: Weekly reflection.

DAY 16 — Monday, 23 November 2026 | LEARNING | 4 HOURS
Topic: Dictionaries and sets
Study: Key-value pairs; lookup; update; iteration; set uniqueness.
Do: Represent a robot configuration as a dictionary with name, wheel diameter and sensor count. Validate required keys.
Save: robot_config.py.

DAY 17 — Tuesday, 24 November 2026 | LEARNING | 4 HOURS
Topic: File handling
Study: open(); with; read/write/append; paths; text files; CSV concept.
Do: Save a mock robot mission log to a text file, read it back and display the records.
Save: session_log.py and generated log file.

DAY 18 — Wednesday, 25 November 2026 | LEARNING | 4 HOURS
Topic: Debugging and exceptions
Study: Syntax/runtime/logic errors; tracebacks; try/except; debugger basics.
Do: Intentionally introduce three bugs into copied exercises; diagnose and fix each; handle invalid numeric input.
Save: debugging_lab.py and bug diary.

DAY 19 — Thursday, 26 November 2026 | LEARNING | 4 HOURS
Topic: Modules and standard library
Study: import; math; pathlib; random; module organization.
Do: Use math.pi to calculate wheel circumference. Move helper functions into a separate file and import them.
Save: robot_math.py and helper module.

DAY 20 — Friday, 27 November 2026 | WEEKLY ASSESSMENT | 4 HOURS
Topic: Object-oriented preview
Study: Classes; objects; attributes; methods; __init__; basic encapsulation.
Do: Create a Sensor class with name, unit and a method to format a sample reading. Create two objects.
Save: sensor_class.py.

DAY 21 — Saturday, 28 November 2026 | LEARNING | 4 HOURS
Topic: Git fundamentals
Study: Repository; working tree; staging; commit; status; log; .gitignore.
Do: Initialize a Git repository for Month1 code. Add .gitignore. Make two meaningful commits and inspect history.
Save: Local Git repository and commit history.

DAY 22 — Sunday, 29 November 2026 | WEEKLY ASSESSMENT | 3.5 HOURS
Topic: Week 3 review
Study: Dictionaries, files, debugging, modules, OOP preview and Git.
Do: Build a small program, debug it and commit the finished version. Explain your changes.
Save: Practical test and Git log.

DAY 23 — Monday, 30 November 2026 | REST / REVIEW | 0 HOURS
Topic: Rest and backup
Study: No new topic. Organize and back up repository.
Do: Optional review; confirm no secrets or passwords are stored in Git.
Save: Clean repository.

DAY 24 — Tuesday, 01 December 2026 | LEARNING | 4 HOURS
Topic: GitHub and remote repositories
Study: Remote; push; pull; README; repository visibility; credential safety.
Do: Create a learning repository. Add README. Push Month1 code and verify files on GitHub. Never commit .env or keys.
Save: GitHub repository with README and source files.

DAY 25 — Wednesday, 02 December 2026 | LEARNING | 4 HOURS
Topic: Linux and terminal fundamentals
Study: pwd, ls, cd, mkdir, touch, cp, mv, rm; relative/absolute paths; Windows PowerShell equivalents.
Do: Navigate a practice folder; create, rename and copy files. Record equivalent PowerShell commands.
Save: terminal_commands.md.

DAY 26 — Thursday, 03 December 2026 | LEARNING | 4 HOURS
Topic: Processes and environments
Study: Processes; PATH; environment variables; virtual environments; package installation concept.
Do: Run a script from terminal. Inspect Python executable path. Create and activate a virtual environment.
Save: environment_notes.md and working venv.

DAY 27 — Friday, 04 December 2026 | PROJECT | 4 HOURS
Topic: Plan the monthly project
Study: Requirements; inputs/process/outputs; pseudocode; validation; test cases.
Do: Specify Robot Mission Logger: mission name, duration, status and notes. Write pseudocode and at least five test cases.
Save: mission_logger_spec.md.

DAY 28 — Saturday, 05 December 2026 | PROJECT | 4.5 HOURS
Topic: Build Robot Mission Logger
Study: Combine variables, input, conditions, loops, functions, dictionaries and file persistence.
Do: Implement CLI menu to add and list missions. Validate fields. Save records locally as CSV or JSON.
Save: Robot Mission Logger v1 source code.

DAY 29 — Sunday, 06 December 2026 | PROJECT | 4.5 HOURS
Topic: Test and document project
Study: Normal/boundary/invalid cases; refactoring; README; limitations.
Do: Run at least ten test scenarios. Fix bugs. Write setup, features, usage and limitations. Commit final version.
Save: Test checklist, README and GitHub commit.

DAY 30 — Monday, 07 December 2026 | WEEKLY ASSESSMENT | 4 HOURS
Topic: Final assessment and demo
Study: Explain core topics; demonstrate project; self-evaluation; next-month readiness.
Do: Take a 25-question fundamentals quiz. Give a five-minute demo. List three strengths and three gaps.
Save: Assessment record, demo evidence and Month 2 readiness note.

5. Month 1 project — Robot Mission Logger v1

Purpose: Build a small command-line program that records mock robot missions. This gives you practice with the exact Python skills covered this month without needing hardware.

• Mission fields: mission name, duration, status and notes.

• Menu: add a mission, list saved missions, and exit safely.

• Validation: reject blank mission names, negative durations and unsupported status values.

• Persistence: save and load records from a local CSV or JSON file.

• Testing: at least ten cases, including normal inputs, empty fields, invalid numbers and missing data file.

• Documentation: README with setup, how to run, features, sample output and known limitations.

• Version control: commit in small meaningful steps and push to your GitHub repository. Never commit passwords, API keys, .env files or private credentials.

6. Month-end pass criteria

■ Can write a Python script from a blank file using variables, conditions, loops and functions.

■ Can use lists and dictionaries to organize data.

■ Can read and write a local text/CSV/JSON file.

■ Can identify and fix basic syntax, runtime and logic errors.

■ Can explain what a Git repository, commit, remote and push do.

■ Can run scripts and basic file commands from PowerShell.

■ Can demonstrate Robot Mission Logger v1 and explain its limitations.

7. What comes next

Month 2 will build on this foundation with C++ fundamentals and electronics: circuit basics, voltage/current/resistance, breadboards, LEDs, resistors and multimeter use. Do not buy a board, motor driver or sensor kit until the Month 2 equipment list is finalized. Your Month 1 assessment should determine whether to move at normal pace or repeat selected Python topics.